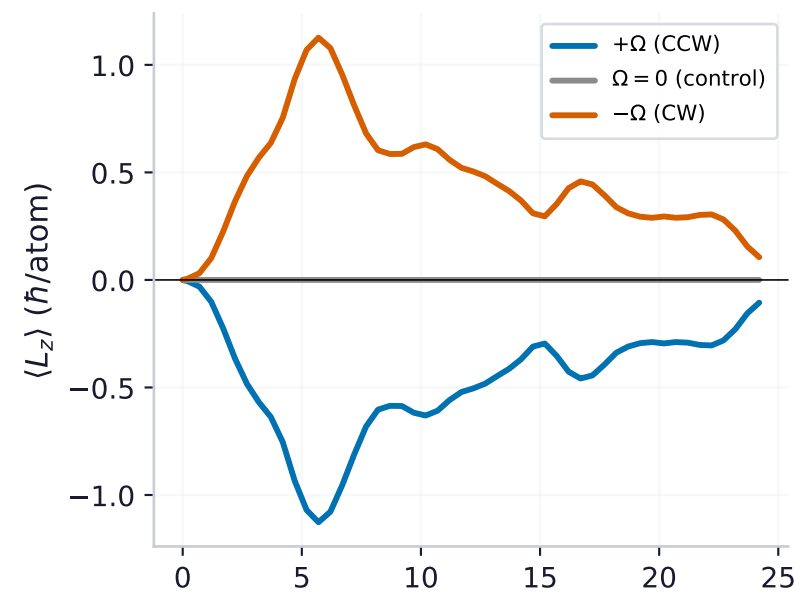
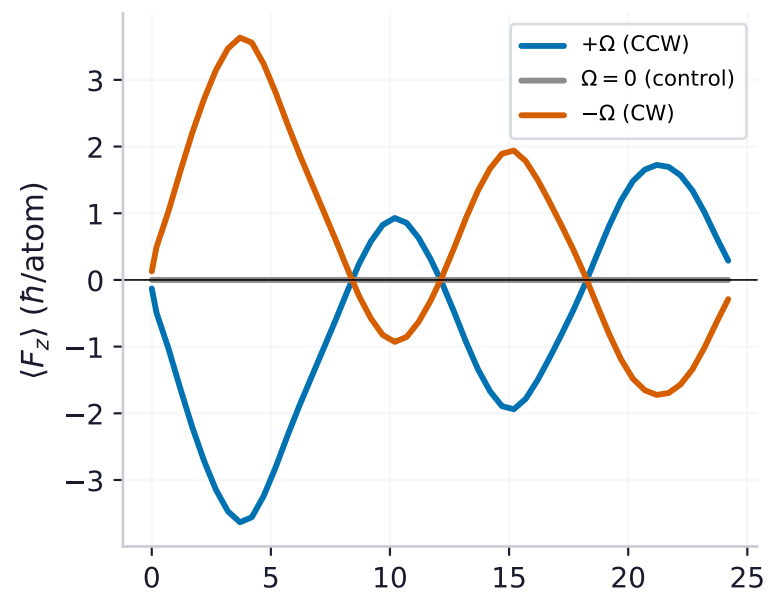


Rotating magnetic field → quantum vortices + Barnett spin excitation, controlled by rotation direction (¹⁵¹Eu F=6 dipolar BEC)

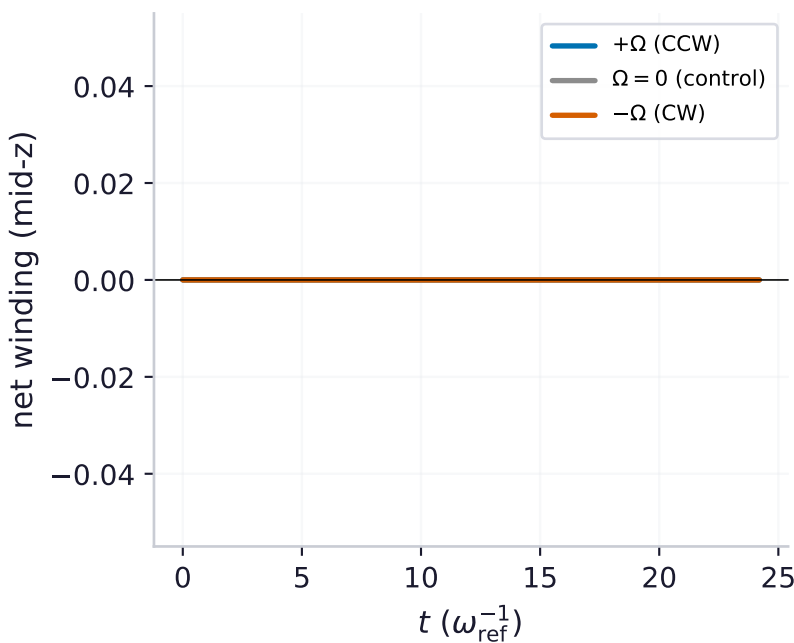
(a) orbital $\langle L_z \rangle$ (vortices)



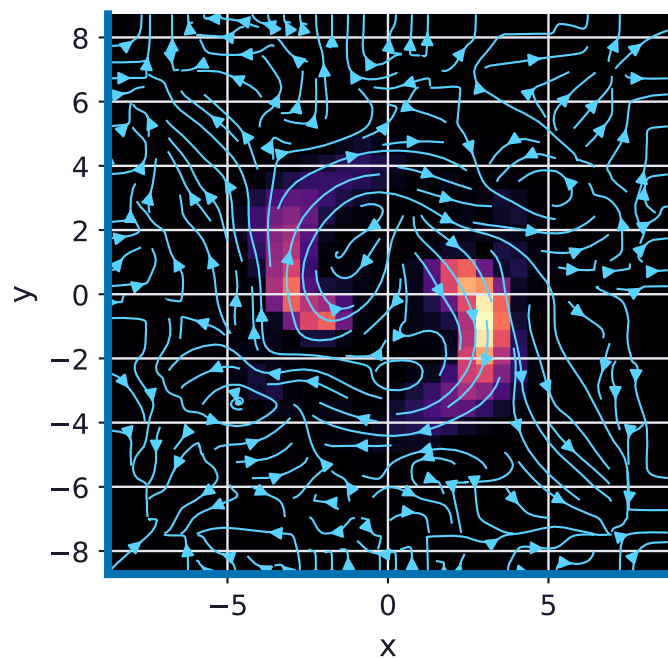
(b) Barnett spin $\langle F_z \rangle$



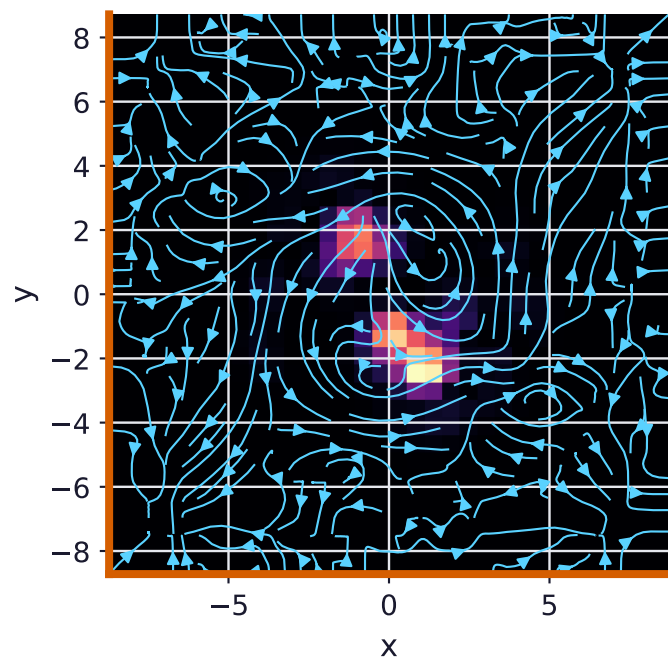
(c) net vortex winding (chirality)



$+\Omega$ (CCW): density + current ($t=7.5$)
circulation reverses with rotation



$-\Omega$ (CW): density + current ($t=7.5$)
circulation reverses with rotation



Rotation-controlled Barnett + vortex
(transverse $J_z=0$ start, unitary)

$\langle L_z \rangle$ time-avg:
 $+\Omega$: -0.46
 $\Omega = 0$: 0.00
 $-\Omega$: +0.46

$\langle F_z \rangle$ peak amplitude: 3.6

MIRROR (residual 0%):
 $L_z(+\Omega) = -L_z(-\Omega)$
 $F_z(+\Omega) = -F_z(-\Omega)$

Vortex chirality: $+\Omega \rightarrow$ net -2,
 $-\Omega \rightarrow$ net +2 (real cores).

$\Omega = 0$ control: nothing.